Unifying Theory of Dynamic Aether Mechanics: Cosmology

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Abstract

Within the Dynamic Aether Mechanics framework, cosmological phenomena emerge from the three independently varying properties of the Aether medium (density, pressure, temperature). This paper explains cosmological redshift as real Doppler shift from thermal expansion of cosmic voids, dark energy as the thermal pressure of hot void Aether, and dark matter as the two-fluid first-sound correction (the outward increase in c_1 from the varying normal fraction provides additional centripetal acceleration through the acoustic metric). Black holes are identified as sonic horizons where Aether inflow velocity reaches c, providing a concrete physical mechanism for Hawking radiation via asymmetric transient binding at the horizon boundary. Neutron stars serve as natural laboratories for extreme-density Aether physics, with pulsar glitches arising from quantized vortex avalanches.

This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [40].

1. Cosmological Effects

This is where the interplay between Aether's independently varying properties—particularly temperature and the two-fluid normal fraction—determines the large-scale structure and behavior of the universe.

Over cosmological timescales, two processes have shaped the thermal state of the Aether:

1. Permanent binding (driven by the Dynamic Casimir Effect in high-energy environments) has created matter, locking thermal energy into mass and cooling the Aether in matter-containing regions.

2. Stars and other luminous matter radiate energy outward into surrounding voids, where no mass formation absorbs it. This accumulated radiation, together with viscous dissipation from the Hubble flow (Section 1.2), heats the Aether in voids far above the temperature near matter.

The Aether density ρ_A is essentially uniform ($\delta\rho_A/\rho_A \sim 10^{-4}$; see Section 1.7), because $K_A \sim 10^{28}$ J/m^3 resists compression far more effectively than any cosmological force can drive it. The cosmologically relevant variable is *temperature*: matter-containing regions have cold, low-normal-fraction Aether, while deep space voids have hot, high-normal-fraction Aether. These two thermal environments produce different large-scale effects.

1.1 Cosmological Redshift

The high temperature and thermal pressure of Aether in voids produces a real, physical outward force on the matter at void boundaries. Since matter regions have low thermal pressure (cold, despite high baryon density), there is a net pressure differential pushing matter apart—causing real expansion of the space between galaxy clusters.

This expansion produces real Doppler redshift. Light from distant galaxies is redshifted because those galaxies are genuinely receding, driven apart by the thermal pressure of hot voids. The greater the distance, the more intervening void expansion the light has traversed, and the greater the cumulative redshift. This produces the distance-proportional redshift described by Hubble's law [12].

Critically, because this is real expansion (not an artifact of energy loss or medium transitions), it naturally produces the time dilation of distant supernova light curves that is observed and predicted by expansion-based cosmology. The light curves are stretched because the space the light travels through is physically expanding during transit—exactly as in standard cosmological models, but driven by Aether thermal pressure rather than residual momentum from an initial explosion.

1.2 Dark Energy

The accelerating expansion of the universe finds a natural explanation in this framework.

As matter clusters more tightly under gravity (consuming Aether and locking energy into mass), it radiates increasing amounts of energy outward into the surrounding voids. Stars emit light, heat, and radiation that propagates into the Aether of deep space. Since voids contain no mass formation to absorb this energy, it accumulates as thermal energy in the Aether, raising the temperature and thermal pressure of the voids. However, the integrated stellar radiation over cosmic history ($\sim 10^{-14}$ J/m^3) falls $\sim 10^4$ times short of the observed dark energy density ($\rho_\Lambda c^2 \sim 6 \times 10^{-10}$ J/m^3). The two-fluid model (Section 1.3) provides the missing energy channel: the normal fraction of the Aether in voids has finite *bulk* viscosity ζ_n and dissipates energy from the cosmic expansion. The Hubble flow $\mathbf{v} = H\mathbf{r}$ is pure dilation ($\nabla \cdot \mathbf{v} = 3H$) with zero shear, so only bulk viscosity contributes. The dissipation rate $\dot{\varepsilon} = \zeta_n (3H)^2 = 9\,\zeta_n H^2$ accumulates over cosmic time to fill the energy budget, requiring a normal-component bulk viscosity

$$\zeta_n \;\approx\; \frac{\rho_\Lambda\, c^2}{9\, H^2\, t_0} \;\sim\; 3\times 10^7 \text{ Pa s} \tag{1}$$

(where t_0 is the age of the universe). This is an upper bound: H was larger in the past, so the time-integrated heating $\int 9\,\zeta_n\, H^2\, dt$ exceeds $9\,\zeta_n\, H_0^2\, t_0$ by a factor of ~30–40, giving a more accurate estimate $\zeta_n \sim 10^6$ Pa s. This value is an *observational constraint* on the Aether's microscopic properties: the observed dark energy density determines what the bulk viscosity must be. In superfluids, bulk viscosity arises from relaxation between the superfluid and normal components (the three-phonon process) and can be substantial near T_c; whether the Aether's excitation spectrum produces $\zeta_n \sim 10^6$ Pa s is a testable prediction that shares the same unknown—the excitation spectrum—as the dark matter magnitude and the Hubble tension temperature ratio (Section 1.3, Section 1.7). Crucially, photons propagate without dissipation because the superfluid component has zero shear viscosity, and direct photon–quasiparticle scattering is suppressed by $P_{\text{th}}/K_A \sim 10^{-38}$. This heating therefore occurs without attenuating electromagnetic radiation—the two-fluid model provides a natural separation between bulk-viscous heating (which operates on the normal fraction) and transparent photon propagation.

This creates a positive feedback loop: matter clusters and radiates energy → voids heat up → thermal pressure increases → expansion accelerates → voids grow larger → more surface area to absorb radiation → voids heat up further.

This naturally explains several observed features of dark energy:

- Why dark energy constitutes the vast majority of the universe's energy budget: the vast majority of space is void, filled with hot Aether.
- Why expansion is accelerating rather than decelerating: the thermal pressure in voids is actively increasing as stars continue to radiate energy into them.
- Why dark energy appears as a property of space itself rather than a localized force: it is a property of the Aether that fills space, driven by the temperature differential between voids and matter regions.

1.3 Dark Matter

Dark matter—the observation that galaxies and galaxy clusters exhibit stronger gravitational effects than their visible mass accounts for—is explained in the Aether framework by the two-fluid first-sound correction: the outward increase in the first-sound speed c_1 from the varying normal fraction provides additional centripetal acceleration through the acoustic metric, with no additional mass required.

The two-fluid Aether. As a quantum superfluid, the Aether obeys the Landau–Tisza two-fluid model: below a critical temperature T_c, the medium consists of two interpenetrating components—a superfluid fraction ρ_s/ρ_A (zero viscosity, carries no entropy) and a normal fraction ρ_n/ρ_A (finite viscosity, carries entropy), with $\rho_s + \rho_n = \rho_A$ [37]. The superfluid fraction varies continuously with temperature: near matter (cold, $T \ll T_c$), $\rho_s/\rho_A \approx 1$; in warmer regions (approaching T_c), the normal fraction grows. Crucially, photons—spin-wave excitations of the condensate (Paper 7 [41])—propagate without absorption or scattering, because direct photon–quasiparticle energy exchange is suppressed by $P_{\rm th}/K_A \sim 10^{-38}$. The universe therefore remains transparent regardless of the local normal fraction. However, the normal fraction *does* modify the background propagation speed: through spin-orbit coupling [36], the photon's microscopic spin-wave character produces macroscopic mass currents indistinguishable from first-sound perturbations (Paper 7, Section 4), so the effective photon speed at macroscopic wavelengths is the two-fluid first-sound speed c_1, not the bare condensate speed c. The viability of superfluid dark matter has been independently demonstrated by Berezhiani and Khoury [37], who showed that a superfluid medium with appropriate phonon dynamics can reproduce MOND phenomenology on galactic scales.

Observational reinterpretation. Every dark matter measurement ultimately observes a gravitational effect and infers mass using standard gravity ($v^2 = GM/r$ or $\theta = 4GM/(rc^2)$). In the Aether framework, the relation is $v^2 = GM_{\rm visible}/r + (r/2)(dc_1^2/dr)$ (Eq. 3), so what is conventionally attributed to "dark matter mass" is actually the first-sound gradient term—no additional mass is required. Crucially, both light and matter follow geodesics of the *same* acoustic metric (light propagates at c_1 through spin-orbit coupling, as described above), so the c_1 gradient that enhances rotation curves automatically produces the corresponding gravitational lensing.

Quantitative mechanism: two-fluid first-sound correction. In the Landau two-fluid model, the first-sound speed—the adiabatic sound speed $c_1^2 = (\partial P/\partial\rho)_S$—is not simply $c = \sqrt{K_A/\rho_A}$ but receives a thermal correction from the normal component [37]:

$$c_1^2 \;=\; c^2 \;+\; \frac{T\,s^2\,\rho_s}{c_v\,\rho_n} \tag{2}$$

where s is the entropy per unit mass, c_v is the specific heat per unit mass, and ρ_s/ρ_n is the superfluid-to-normal density ratio. At $T = 0$ there are no excitations and trivially $c_1 = c$. At any finite temperature the correction is *positive* ($c_1 > c$); for a pure phonon gas the correction is $c^2/3$ (temperature-independent), while for a general excitation spectrum the correction varies with temperature—and it is this temperature-dependent part that produces the dark matter effect through its spatial gradient.

Why photons propagate at c_1. Photons are microscopically spin-wave excitations of the spinor condensate—transverse oscillations that do not change the local density (Paper 7 [41]). However, through spin-orbit coupling [36], these spin oscillations produce effective mass currents that are indistinguishable from velocity-field (density/flow) perturbations at macroscopic scales. The crossover occurs at the spin healing length $\xi_s \lesssim 10^{-27}$ m—all observable light is deeply in the macroscopic regime, where the effective propagation speed is the first-sound speed c_1. Transparency is maintained because direct photon–quasiparticle energy exchange (absorption, scattering) is suppressed by $P_{\mathrm{th}}/K_A \sim 10^{-38}$—a measure of scattering cross-section, not of the background speed modification. The c_1 correction is purely thermodynamic and therefore wavelength-independent: gravitational lensing from the dark matter component is achromatic, consistent with observations.

Since the normal fraction increases outward from the galaxy (warmer Aether has more thermal excitations), c_1 *increases outward*. Extending the acoustic metric of Paper 3 (Section 2.1) to position-dependent wave speed $c_1(r)$, the circular-orbit condition $v_{\mathrm{orbit}}^2 = (r/2)\, dg_{tt}/dr$ gives:

$$v_{\mathrm{orbit}}^2 = \frac{GM}{r} + \frac{r}{2}\frac{dc_1^2}{dr} \tag{3}$$

the positive gradient $dc_1^2/dr > 0$ provides additional centripetal acceleration—precisely the "dark matter" effect. The sign is *automatically correct*: it follows from the Landau two-fluid thermodynamics without any assumptions about which phase is stiffer. The effect is naturally cored: in the pure superfluid center ($\rho_n \approx 0$), the correction vanishes and only visible-mass gravity operates.

Magnitude of the first-sound correction. The dark matter effect depends on the *spatial variation* Δc_1^2 across the halo, which requires the correction to vary with temperature. The magnitude of this variation depends critically on the Aether's entropy per unit mass s, which in turn depends on the full excitation spectrum (phonons, and any roton-like or other modes beyond the long-wavelength limit). Two limiting cases bracket the answer:

- **Phonon gas only** (long-wavelength collective modes): For a pure phonon gas ($\omega = ck$), the correction $Ts^2\rho_s/(c_v\rho_n) = c^2/3$ is *temperature-independent* (the phonon scaling $s \propto T^3$, $\rho_n \propto T^4$ produces exact cancellation). A constant correction has zero spatial gradient and produces no dark matter effect; the residual variation from dispersion corrections is $\Delta c_1^2/c^2 \sim 10^{-14}$—negligibly small.
- **Ideal BEC** (all single-particle excitations): $\Delta c_1^2/c^2 \sim 10^{-4}$—∼50× larger than needed.

The observed dark matter requires $\Delta c_1^2/c^2 \sim 10^{-6}$ across the halo, corresponding to an entropy intermediate between these limits. For comparison, helium-4 at $T/T_c \approx 0.7$ has entropy dominated by roton excitations, placing it near the ideal-BEC limit; whether the Aether has analogous intermediate-wavelength excitations is the key open question.

Quantitative consistency across observations. A flat rotation curve ($v_{\rm flat}$ = const) requires $dc_1^2/dr = 2v_{\rm flat}^2/r$, giving a logarithmic profile $\Delta c_1^2(r) = 2v_{\rm flat}^2 \ln(r/r_0)$. The total variation across a Milky-Way-sized halo ($v_{\rm flat}$ = 220 km/s, 1–200 kpc) is $\Delta c_1^2/c^2 \approx 6 \times 10^{-6}$—six parts per million, undetectable by any terrestrial experiment.

Gravitational lensing. Because both light and matter follow geodesics of the same acoustic metric, the effective mass profile from the c_1 gradient is exactly the singular isothermal sphere (SIS): $M_{\rm eff}(r) = v_{\rm flat}^2 r/G$. The corresponding deflection angle is $\theta = 2\pi v_{\rm flat}^2/c^2 \approx 0.70''$, identical to the SIS result $\theta_{\rm SIS} = 4\pi\sigma^2/c^2$ (with $\sigma = v_{\rm flat}/\sqrt{2}$). No separate lensing calculation is needed—rotation curves and lensing are automatically consistent.

Galaxy clusters. For a massive cluster ($\sigma \sim 1000$ km/s), the required $\Delta c_1^2/c^2 \approx 10^{-4}$—roughly 17× larger than for a galaxy. This is physically expected: clusters are hotter (more normal fraction, higher entropy), so the first-sound correction is naturally larger.

Bullet Cluster. The c_1 correction is maintained by each galaxy's cooling of the surrounding Aether through the binding process. During the collision, the shocked gas heats the local Aether, reducing the normal-fraction contrast and weakening the c_1 correction in the collision region. The surviving corrections follow the galaxies, reproducing the observed lensing–gas separation.

No dark matter particles. The mechanism is a thermodynamic property of the two-fluid medium, not a new particle species, explaining the null results of direct-detection experiments.

Tully–Fisher relation. The observed baryonic relation $M \propto v^4$ constrains the model. For steady-state thermal conduction, $T_{\rm void} - T(R_{25}) = L_{\rm cool}/(4\pi\kappa R_{25})$, and the orbital velocity at R_{25} is $v^2 \propto (L_{\rm cool}/R_{25})^{1+\beta}$, where β parametrizes the critical behavior of the first-sound correction near T_c: $f'(T) \propto (T_{\rm void} - T)^\beta$. The Tully–Fisher slope depends on *both* β and how the galaxy's cooling rate scales with mass ($L_{\rm cool} \propto M^\alpha$). For $\alpha = 1$ (linear cooling) and $\beta = 0$ (smooth, ideal BEC), the model gives $M \propto v^3$ (wrong slope). However, $\beta = -1/4$ (a weak cusp in $f'(T)$, as seen in helium-4's lambda anomaly) yields $M \propto v^4$ exactly. Alternatively, sublinear cooling ($\alpha \approx 0.7$, as expected from the observed star-formation main sequence) with smooth $\beta \approx +0.35$ also yields $M \propto v^4$. The correct slope thus depends on the same excitation spectrum and critical behavior that determine the dark matter magnitude—an open but specific calculation.

Quantitative status and open calculations. The first-sound correction mechanism has several structural advantages: (a) the sign is correct by construction (Landau thermodynamics); (b) cored profiles emerge naturally (no correction in the pure superfluid center); (c) it is independent of the quantum of circulation κ_0; (d) rotation curves and lensing are automatically consistent (same acoustic metric); (e) the scaling from galaxies to clusters is qualitatively correct. The critical open calculations are: (i) the Aether's excitation spectrum beyond the phonon regime, which determines the entropy and hence the magnitude ($\Delta c_1^2/c^2$ must be $\sim 10^{-6}$ for galaxies; the phonon-only limit gives 10^{-14} and the ideal BEC gives 10^{-4}); (ii) the three-dimensional temperature profile $T(r, z)$

from first principles; (iii) the Tully–Fisher exponent (currently ~3 vs. observed ~4); (iv) the CMB power spectrum from the cosmological superfluid phase transition; (v) whether the c_1 correction propagates differently from collisionless dark matter during acoustic oscillations. The viability of superfluid dark matter has been demonstrated in principle by Berezhiani and Khoury [37] through phonon-mediated forces; the Aether's two-fluid first-sound mechanism operates through a different channel but within the same superfluid dark matter paradigm.

1.4 Cosmic Microwave Background Radiation

In this theory, photons are spin-wave excitations of the Aether condensate (Paper 7 [41]). The cosmic microwave background is therefore not radiation traveling *through* the medium but the thermal excitation spectrum *of* the medium—the Aether at thermal equilibrium. The CMB temperature ($T_{\mathrm{CMB}} = 2.725$ K) is the Aether's temperature, and the near-perfect blackbody spectrum confirmed by COBE/FIRAS [42] follows naturally: it is the equilibrium spectrum of a quantum medium.

The CMB anisotropies ($\delta T/T \sim 10^{-5}$) reflect spatial temperature variations in the Aether. The specific angular power spectrum—with its characteristic acoustic peaks encoding baryon–photon physics—remains an open question within the Aether framework. Two candidate mechanisms are identified in the Open Questions section: the Kibble–Zurek defect spectrum from the cosmological superfluid phase transition, and second-sound oscillations in the two-fluid Aether. Deriving the CMB power spectrum from first principles is one of the critical open calculations for this theory.

This theory does not preclude the Big Bang; it offers complementary mechanisms for phenomena traditionally attributed solely to it.

1.5 Observational Evidence for Dynamic Dark Energy

The thermal pressure mechanism makes a distinctive prediction: dark energy should *evolve over time*. As voids expand and accumulate radiative energy, their thermal pressure changes—it is not a fixed cosmological constant but a dynamic quantity driven by the ongoing thermodynamic interplay between matter-containing regions and voids. This contrasts sharply with the standard ΛCDM model, in which the cosmological constant Λ is strictly time-independent.

Recent observations strongly favor the dynamic prediction. The Dark Energy Spectroscopic Instrument (DESI) collaboration, using baryon acoustic oscillation measurements from over 14 million galaxies and quasars [14], has found evidence at 2.8–4.2σ significance that the dark energy equation of state parameter w varies with redshift. The data prefer a model in which the dark energy density is *slowly decreasing* with time, inconsistent with a cosmological constant at the ~3σ level when combined with cosmic microwave background and supernovae data.

This is precisely what the Aether thermal pressure model predicts. Two competing effects govern the evolution:

- **Heating**: Stars continuously radiate energy into voids, raising the thermal pressure. This drives the positive feedback loop described above.
- **Dilution**: As voids expand, the Aether within them becomes less dense. Since pressure depends on both temperature and density, the net effect on expansion rate depends on the balance between heating and dilution.

The observed decrease in dark energy density is consistent with dilution beginning to dominate over heating at late cosmological times—a natural outcome when the rate of void expansion outpaces the rate of energy injection from matter regions.

An independent line of evidence comes from the Finsler gravity framework [16], which extends general relativity to incorporate the thermal pressure of kinetic gases directly into the field equations. When this is done, the Friedmann equations naturally produce cosmic acceleration without requiring a separate dark energy term—the acceleration arises from the thermal pressure of the medium itself. While the mathematical formalism differs from the Aether model, the physical mechanism is strikingly parallel: in both cases, cosmic acceleration is driven by the thermal pressure of a space-filling medium rather than by a cosmological constant.

The Dark Energy Survey's final analysis [15], combining weak lensing and galaxy distribution measurements over 6 billion years of cosmic history, provides the tightest constraints yet on the dark energy equation of state—and these constraints are consistent with a slowly varying, dynamic dark energy rather than a strict cosmological constant.

1.6 Void Structure and Differential Expansion

The Aether framework predicts that cosmic voids should exhibit specific, universal properties dictated by the interplay of density, pressure, and temperature. In particular, since K_A is invariant and ρ_A varies smoothly, the void density profile should be governed by the same physics everywhere—producing similar profiles across different void sizes and cosmic epochs.

Galaxy surveys confirm this prediction. When void density profiles are scaled by the void effective radius, the resulting profiles are approximately universal: the shape is largely independent of void size and redshift. This universality is difficult to explain if voids are shaped purely by initial conditions and gravitational dynamics (which would produce diverse profiles depending on local environment and formation history), but follows naturally from a medium whose bulk modulus K_A is the same everywhere.

Furthermore, galaxies within voids show systematically different properties from those in denser environments: lower stellar masses, bluer colors, later morphological types, higher gas fractions, and lower metallicities. These "delayed evolution" signatures are consistent with the higher Aether temperature in voids: the stronger first-sound correction (larger normal fraction) shifts the effective gravitational dynamics, and the higher thermal energy potentially raises thresholds for permanent binding (the DCE threshold depends on local conditions).

The timescape cosmology program [17] has demonstrated that differential expansion rates between voids and walls—exactly the behavior predicted by the Aether three-property framework—can account for the observed distance-redshift relation of Type Ia supernovae without requiring dark energy at all. In that model, as in the Aether model, voids expand faster than matter-containing regions because they contain less gravitational deceleration. The Aether framework adds a physical mechanism: the thermal pressure differential between hot voids and cold matter regions actively drives the differential expansion.

1.7 The Hubble Tension

One of the most pressing open problems in modern cosmology is the *Hubble tension*: a persistent, statistically significant discrepancy between early-universe and late-universe measurements of the expansion rate H_0. Observations of the cosmic microwave background by the Planck satellite, analyzed within ΛCDM, yield $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} [32]. Local distance-ladder measurements using Cepheid-calibrated Type Ia supernovae give $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} [33]—a discrepancy of ∼8%, exceeding 5σ significance. Independent measurements from gravitational lensing time delays also favor the higher value [34], and the James Webb Space Telescope has ruled out Cepheid crowding artifacts at $>8\sigma$ confidence. The tension is increasingly regarded as evidence that the standard cosmological model is incomplete.

The Aether framework addresses this tension through spatially varying dark energy: because the thermal pressure that drives cosmic expansion depends on local temperature, different environments expand at different rates.

The local void. Observational surveys have established that our galaxy resides within the Keenan–Barger–Cowie (KBC) void [35], a region approximately 300 Mpc in radius. The observed luminosity density contrast is $\delta \equiv 1 - \rho/\bar{\rho} \approx 0.46 \pm 0.06$, indicating that the local universe is roughly 46% less dense in *matter* than the cosmic average. Approximately half of this apparent contrast is an artifact of void-induced outflows that distort the redshift–distance relation; the physical matter underdensity is estimated at ∼20–30%.

In ΛCDM, the KBC void poses a severe challenge: a void of this size and depth is expected fewer than 1 in 10^9 times in standard simulations [35]. The Aether framework resolves this problem. The thermal pressure differential between hot void interiors and cold matter-containing walls (Section 1.2) actively creates and sustains large voids—the KBC void is not a statistical fluke but the natural outcome of the Aether's three-property dynamics.

Why the standard void outflow alone does not resolve the tension. In an underdense region, matter experiences a net outward peculiar velocity driven by the gravitational pull of the denser surrounding walls. In linear perturbation theory, the peculiar velocity within a spherical underdensity of contrast δ is:

$$v_{\rm pec}(r) \;\approx\; -\frac{1}{3}\,H_0\,f(\Omega_m)\,\delta\,r \tag{4}$$

where $f \approx \Omega_m^{0.55} \approx 0.53$. For a physical underdensity of ~25%, this outflow contributes $\Delta H_0 \approx +3\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. However, the void's matter deficit also *reduces* the local expansion rate through the Friedmann equation: with less matter contributing to the total energy density, $H^2_{\rm local} \propto \Omega_m(1-\delta)+\Omega_\Lambda$ is lower than the cosmic average, contributing $\sim -2.6\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. These two effects nearly cancel, leaving a net boost of only $\sim 0.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$—far short of the observed $+5.6\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ discrepancy. This cancellation is well established: analyses within standard ΛCDM have shown that local voids cannot resolve the Hubble tension when dark energy is spatially uniform.

Resolution through temperature-dependent dark energy. The Aether model breaks this cancellation. In ΛCDM, the cosmological constant Λ is spatially uniform—voids and walls experience the same dark energy density. In the Aether model, dark energy is the thermal pressure of the Aether, which depends on local *temperature* (Section 1.2). Crucially, the Aether density ρ_A is essentially uniform across cosmological structures: because $K_A \sim 10^{28}\ \mathrm{J/m^3}$ is enormous, the Aether resists compression and rarefaction far more effectively than any cosmological force can drive it. The gravitational density perturbation from the KBC void is only $\delta\rho_A/\rho_A \sim 2G\Delta M/(Rc^2) \sim 10^{-4}$, so the speed of light $c = \sqrt{K_A/\rho_A}$ is effectively constant across cosmic structures (varying by less than 0.01%). Similarly, although $G \propto \epsilon\,\rho_A\,\psi^2$ is in principle spatially variable (Section 1.3), both channels of variation are negligible here: the density channel contributes $\delta G/G \sim \delta\rho_A/\rho_A \sim 10^{-4}$, and the temperature channel (through the asymmetry fraction ϵ) is suppressed by the ratio $P_{\rm th}/K_A \sim 10^{-37}$—the thermal energy is far too small relative to the medium's mechanical energy scale to perturb the bind–unbind cycle. The Friedmann equation below therefore uses constant G. (Intriguingly, at the CMB temperature this ratio drops to $\sim 10^{-43}$, numerically equal to the asymmetry fraction ϵ itself; Paper 3 [38] explores the speculative possibility that the two are physically identified through the normal-fluid fraction of the two-fluid model, yielding $T \approx T_{\rm CMB}$ for $\rho_A \approx 1.3 \times 10^{12}\ \mathrm{kg/m^3}$.)

Temperature, by contrast, varies dramatically. Voids accumulate radiative energy with no mass formation to absorb it, reaching temperatures far above matter-containing regions. Since the thermal pressure depends on temperature, the effective dark energy is higher in voids and lower near matter.

This modifies the local Friedmann equation. Define η as the fractional thermal pressure excess of the void above the cosmic average, so that the effective dark energy density in the void is $\Omega_\Lambda(1+\eta)$.

The locally measured H_0 then has two contributions:

$$H_{0,\text{local}} = \underbrace{H_{0,\text{avg}} \sqrt{\Omega_m (1-\delta) + \Omega_\Lambda (1+\eta)}}_{\text{local Friedmann rate}} + \underbrace{\tfrac{1}{3} H_{0,\text{avg}} \, f \, \delta}_{\text{peculiar velocity}} \tag{5}$$

The first term is the local expansion rate (modified by both the matter deficit and the thermal pressure excess); the second is the gravitational outflow from the void density gradient.

For $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $\delta = 0.25$, matching the observed $H_{0,\text{local}} = 73.0 \text{ km s}^{-1} \text{Mpc}^{-1}$ requires:

$$\eta \approx 0.22 \tag{6}$$

That is, the KBC void's thermal pressure must be approximately 22% above the cosmic average. This is the analysis's single fitted parameter: Ω_m, Ω_Λ, δ, f, and $H_{0,\text{avg}}$ are all externally measured, and the peculiar velocity and Friedmann deficit are derived from standard physics with no free parameters; only η is determined by matching the observed local expansion rate. The result is robust: varying δ across the plausible range 0.15–0.30 changes η by less than ± 0.01.

If voids occupy a fraction $f_v \approx 0.7$ of the cosmic volume and thermal energy is conserved, the void and wall temperatures satisfy:

$$\frac{T_\text{void}}{T_\text{wall}} \approx 2.5 \tag{7}$$

This ratio assumes the thermal pressure is proportional to temperature ($P_\text{th} \propto T$, as for an ideal gas); a radiation-dominated pressure ($P \propto T^4$) would give a smaller ratio of ~ 1.3. A factor-of-2.5 temperature contrast is physically plausible: voids accumulate radiative energy over cosmological timescales with no mass formation to absorb it, while matter-containing walls continuously lock thermal energy into mass through the binding mechanism (Section 1.2). Whether the Aether's thermodynamic history quantitatively produces this ratio is an open question (see below). Notably, the pressure law ($P \propto T^n$) depends on the Aether's excitation spectrum—the same unknown that determines the dark matter magnitude (Section 1.3), the bulk viscosity ζ_n (Section 1.2), and the dark energy equation of state $w(z)$. This is a powerful consistency test: a single excitation spectrum must simultaneously produce the correct dark matter halo profiles ($\Delta c_1^2/c^2 \sim 10^{-6}$), the correct bulk viscosity for the dark energy density ($\zeta_n \sim 10^6$ Pa s), the correct void temperature ratio for the Hubble tension ($T_\text{void}/T_\text{wall} \approx 2.5$), and the correct $w(z)$ evolution for DESI. If any one of these constraints is incompatible with the others, the theory fails.

The decomposition of effects illustrates why the Aether-specific contribution is essential:

Effect	ΔH_0	Source
Matter deficit (Friedmann)	-2.6	Less matter $\to$ lower H^2
Peculiar velocity outflow	$+3.0$	Gravitational outflow from void
Net without Aether	$+0.4$	*Nearly cancels*
Thermal pressure excess ($\eta \approx 0.22$)	$+5.2$	Hotter void $\to$ higher H^2
Total	$+5.6$	Matches observed tension

The thermal pressure enhancement does essentially all of the work. The standard void outflow ($+3.0$ km s^{-1} Mpc^{-1}) is almost exactly canceled by the Friedmann deficit (-2.6 km s^{-1} Mpc^{-1}), leaving the spatially varying dark energy as the dominant mechanism. The timescape cosmology program [17], which demonstrates that differential expansion between voids and walls can account for Type Ia supernova data, provides independent support for this class of environment-dependent expansion.

Evolving dark energy: a confirmed prediction that does not directly resolve the tension. The DESI collaboration has found evidence at 2.8–4.2σ significance [14] that the dark energy equation of state evolves with redshift, with best-fit parameters indicating a phantom-to-quintessence crossing around $z \approx 0.5$. The Aether thermal pressure model predicts precisely this behavior: at early times, voids were smaller and denser with high thermal pressure relative to their volume (effective $w < -1$); at late times, dilution of expanding void Aether dominates over heating from stellar radiation, producing a decreasing dark energy density ($w > -1$).

This is an important confirmation of the Aether dark energy model. However, the specific phantom crossing favored by DESI does *not* help resolve the Hubble tension. When DESI and Planck data are jointly fit with $w_0 w_a$CDM, the inferred H_0 shifts by less than 0.3 km s^{-1} Mpc^{-1} relative to ΛCDM—leaving the tension essentially unchanged. The physical reason is that the phantom-to-quintessence transition implies decreasing dark energy density at low redshift, which *reduces* late-time acceleration.

In the Aether framework, this is consistent: the Hubble tension arises from *local* temperature variations (the void is hotter), not from the *global* expansion history. The evolving dark energy and the local void are independent features of the same three-property Aether model, and the DESI confirmation validates the dark energy prediction on its own merits.

Testable predictions. The Aether framework for the Hubble tension makes several predictions that distinguish it from other proposed solutions:

1. **Directional anisotropy**: Since the KBC void is not perfectly spherical, the locally measured H_0 should vary across the sky. The Aether model predicts the anisotropy pattern traces the

temperature profile of the void (which determines the thermal pressure), not merely the matter density profile.

2. **Void temperature–expansion correlation**: If thermal pressure drives the differential expansion, there should be a measurable correlation between the thermal energy content of individual voids (inferred from their ISW imprints or SZ signals) and the local expansion rate measured within or behind those voids.

3. **Convergence scale**: The tension should diminish as measurements average over larger volumes that encompass both voids and walls. The Aether model predicts convergence at the scale of statistical homogeneity (~100 Mpc), with a specific functional form governed by the void temperature distribution.

4. **Void abundance**: The Aether model predicts that large voids are more common than ΛCDM simulations allow, because thermal pressure actively sustains them. Upcoming surveys (DESI, Euclid) will map the void size distribution with sufficient precision to test whether ΛCDM's predicted void abundance matches observations.

1.8 The CMB Cold Spot and Supervoid Anomalies

The CMB Cold Spot—a ~5° radius feature approximately 150 μK colder than the mean CMB temperature in the direction of the constellation Eridanus—has remained unexplained since its detection [18]. A supervoid approximately 1.8 billion light-years across has been confirmed in the same direction [19], but under ΛCDM, the integrated Sachs-Wolfe (ISW) effect from this void accounts for only 10–20% of the observed temperature depression.

More broadly, surveys of supervoids elsewhere in the sky have found ISW temperature imprints approximately 5.2 ± 1.6 times stronger than ΛCDM predicts [20]—a significant anomaly that suggests the standard model underestimates the influence of voids on CMB photons.

The Aether framework offers a potential resolution. In standard cosmology, the ISW effect depends only on the time derivative of the gravitational potential as photons traverse a void. In the Aether model, photons traversing a void also encounter different medium properties:

- **Modified flow environment**: Photons traversing a void encounter an outward-expanding Aether flow (driven by the thermal pressure excess $\eta \approx 0.22$; Section 1.7), which modifies their energy through the acoustic metric differently than a uniform Λ would.

- **Higher temperature / lower ϵ**: The hot void Aether has a lower gravitational asymmetry fraction ϵ, weakening the gravitational potential more rapidly as the void expands—amplifying the ISW time derivative $\partial\Phi/\partial t$.

- **Expansion asymmetry**: The void expands faster than the surrounding regions (enhanced thermal pressure), so photons lose more energy climbing out of the expanding void than they gained falling in—but the magnitude depends on the Aether temperature profile, not just the gravitational potential.

The net effect could amplify the ISW signal beyond the ΛCDM prediction, potentially explaining the factor-of-five discrepancy observed in supervoid temperature imprints. The thermal pressure excess $\eta \approx 0.22$ from the Hubble tension analysis (Eq. 6) provides a concrete starting point: the modified Friedmann driving term amplifies $\partial\Phi/\partial t$ by a factor of $\sim$1.6, with the temperature-dependent ϵ and expansion asymmetry contributing additional amplification. A full quantitative calculation remains to be completed (see Open Questions).

2. Black Holes

The event horizon, derived in the gravitational lensing section as a sonic horizon where the Aether inflow velocity reaches c, provides the foundation for a complete Aether-based treatment of black hole physics. Several observed and predicted properties of black holes emerge naturally from the inflow model.

2.1 Hawking Radiation

In 1974, Hawking [1, 2] predicted that black holes are not perfectly black—they emit thermal radiation at a temperature inversely proportional to their mass:

$$T_{\text{Hawking}} = \frac{\hbar\, c^3}{8\pi\, G\, M\, k_B} \tag{8}$$

This prediction has not been directly observed (the radiation is far too weak for stellar-mass black holes), but it is widely accepted on theoretical grounds and has been confirmed in acoustic analogue systems [3].

In the Aether model, Hawking radiation has a concrete physical mechanism rooted in the transient binding cycle.

Far from any mass, the binding cycle is nearly symmetric: Aether binds into a transient pair, the pair annihilates, and the restoring pressure pulse propagates outward. The net energy loss is only the tiny asymmetry ϵ—the gravitational wave.

Near the event horizon, this symmetry breaks down dramatically. The Aether is flowing inward at nearly c, so the restoring pressure pulse from pair annihilation—which propagates outward at the local speed of light—is severely impeded by the inflow. The outward-propagating pulse is redshifted, weakened, and partially swept back inward. The fraction of each cycle's energy that escapes outward is no longer the tiny ϵ of normal-space gravity, but a significantly larger fraction determined by the inflow velocity gradient at the horizon.

Just outside the horizon, some of each cycle's restoring energy still escapes. Just inside, none does—the inflow exceeds c and all restoring pulses are swept inward. The sharp transition at the horizon means that transient binding events straddling the horizon produce a net outward energy flux: the portion of the restoring pulse that forms outside the horizon escapes, while the portion that forms inside is lost. This asymmetry is the physical mechanism of Hawking radiation.

The temperature of this radiation depends on the gradient of the inflow velocity at the horizon—steeper gradients produce sharper asymmetry and hotter radiation. The inflow velocity gradient at the Schwarzschild radius is:

$$\left|\frac{dv_{\rm in}}{dr}\right|_{r_s} = \frac{c^3}{4GM} \tag{9}$$

This is proportional to $1/M$: smaller black holes have steeper gradients and therefore higher Hawking temperatures. This is consistent with Hawking's prediction that $T \propto 1/M$.

The Aether mechanism also connects Hawking radiation to the Unruh effect (identified as an open question elsewhere in this paper). An observer accelerating through Aether at acceleration a experiences the same binding asymmetry as an observer hovering near a horizon with surface gravity a—the acceleration creates an effective inflow gradient that disrupts the symmetry of the local binding cycle. In the Aether model, the Hawking effect and the Unruh effect are the same phenomenon: asymmetric transient binding caused by velocity gradients in the medium, whether those gradients arise from gravitational inflow or from the observer's acceleration.

This identification has a significant theoretical consequence. The acoustic analogue of Hawking radiation has been experimentally confirmed in Bose–Einstein condensate systems [3], where a flowing superfluid creates a sonic horizon and emits thermal phonons at the predicted temperature. In the Aether model, this is not merely an analogy—it is the same physical process occurring in a different medium. The BEC experiment confirms that sonic horizons in flowing media produce thermal radiation, which is exactly what the Aether model predicts for gravitational event horizons.

2.2 The Photon Sphere and Black Hole Shadow

In general relativity, photons can orbit a non-rotating black hole at a radius of $r_{\rm photon} = 3GM/c^2$, equal to 1.5 times the Schwarzschild radius. This photon sphere defines the apparent size of the black hole shadow—the dark silhouette observed by the Event Horizon Telescope in its images of M87* [5] and Sagittarius A* [6].

In the Aether model, the photon sphere arises from the same two mechanisms that produce gravitational lensing: the acoustic metric of the Aether inflow (transverse advection plus time-delay asymmetry). At the photon sphere radius, the combined inward bending from these two effects is strong enough to bend light into a circular orbit. The weak-field bending formula $4GM/(r_0c^2)$ applies to rays passing far from the mass; near the photon sphere, the strong-field regime requires the full density and inflow profiles to be solved self-consistently.

The Event Horizon Telescope measurements of the M87* shadow are consistent with the general-relativistic prediction to approximately 10% precision. Since the Aether model reproduces the weak-field bending angle exactly ($4GM/(r_0c^2)$), the strong-field extension is expected to yield the correct photon sphere radius and shadow size, but this requires quantitative verification through numerical ray-tracing in the Aether density and inflow profiles.

2.3 Frame Dragging

A rotating mass drags the surrounding spacetime into co-rotation—an effect predicted by general relativity (the Lense–Thirring effect [11]) and confirmed by the Gravity Probe B experiment [4] (2011), which measured the frame-dragging precession of gyroscopes in Earth orbit to be 37.2 ± 7.2 milliarcseconds per year, consistent with the predicted 39.2 milliarcseconds per year.

The quantitative derivation of frame dragging from the v^2/c^2 universality is given in Section 1.7.

In the Aether model, frame dragging is a natural consequence of the vortex mechanism. Each electron produces a spin vortex that circulates Aether around its spin axis. In a rotating body, the aggregate vortex circulation of all its electrons produces a net azimuthal Aether flow around the body. This is directly analogous to a rotating body in a viscous fluid dragging the fluid into co-rotation—except that the coupling mechanism is the electromagnetic vortex interaction rather than viscous friction.

For a rotating black hole, the frame-dragging effect creates an ergosphere—a region outside the event horizon where the Aether is dragged so rapidly that no object can remain stationary relative to the distant Aether. Within the ergosphere, all matter and light must co-rotate with the inflowing, rotating Aether. The Penrose process [10] (extracting energy from a rotating black hole) has a natural interpretation: an object entering the ergosphere can exchange angular momentum with the rotating Aether flow, gaining energy at the expense of the black hole's rotational kinetic energy.

2.4 The No-Hair Theorem

In general relativity, black holes are characterized by only three externally measurable quantities: mass, angular momentum (spin), and electric charge. All other information about the matter that formed the black hole is lost behind the event horizon.

The Aether model provides a physical explanation for this result. Inside the event horizon, the Aether inflow exceeds c, sweeping all internal structure inward. The only properties that can influence the external Aether flow pattern are:

- **Mass:** determines the inflow velocity profile $v_{\text{in}}(r) = \sqrt{2GM/r}$ and therefore the inflow velocity profile, time dilation, and gravitational wave output.

- **Angular momentum:** determines the azimuthal Aether flow pattern (frame dragging). The net spin of the infalling matter is preserved as rotational motion of the Aether inflow.

- **Charge:** determines the net coherent axial-oscillation wave field emitted by the bound-core structure of the infalling matter (electric field; Paper 4 [39]). A net positive or negative charge produces a persistent long-range wave field (high-pressure-leading for positive net charge, low-pressure-leading for negative), superimposed on the gravitational inflow.

All other properties of the infalling matter—its composition, temperature, shape, internal structure—are swept behind the sonic horizon and cannot influence the external Aether flow. The no-hair theorem is a consequence of the sonic horizon's one-way nature: only quantities that produce long-range Aether flow patterns (gravity, rotation, electromagnetic field) survive.

2.5 Interior Structure and the Singularity Question

General relativity predicts that all matter inside a black hole collapses to a singularity—a point of infinite density. This prediction is widely regarded as a sign that GR breaks down at extreme scales, requiring a theory of quantum gravity for resolution.

The Aether model offers a different perspective. Inside the event horizon, the Aether inflow exceeds c, and the restoring pressure pulses from transient binding can no longer propagate outward. This has several consequences:

- The transient binding cycle continues (pair formation and annihilation are local processes), but the gravitational wave energy that would normally escape as the tiny asymmetry ϵ is instead carried further inward by the flow. This represents a net inward energy transport that does not occur outside the horizon.

- As Aether accumulates toward the center, the density increases to extreme values. At some density, the perfect linearity of the Aether's pressure–density relationship (K_A = constant) may break down. If K_A increases with extreme compression (the medium stiffens), the inflow would encounter increasing resistance, potentially reaching an equilibrium at finite density rather than collapsing to infinite density.

- At sufficiently extreme density, the binding threshold may be trivially exceeded everywhere, converting all Aether to bound mass. The center of a black hole may contain a core of maximally compressed bound Aether—a state fundamentally different from the mathematical singularity predicted by GR.

- The Planck density ($\rho_{\text{Planck}} = c^5/(\hbar G^2) \approx 5.16 \times 10^{96}$ kg/m^3) represents the scale at which quantum gravitational effects are expected to dominate. In the Aether model, this may correspond to the density at which K_A nonlinearity becomes significant, naturally limiting the compression and avoiding the singularity.

Whether the interior reaches a finite equilibrium or undergoes continued collapse depends on the equation of state of Aether at extreme compression—a regime far beyond current experimental access but potentially constrained by consistency requirements of the theory.

3. Neutron Stars as Aether Laboratories

Neutron stars—collapsed stellar cores with masses of ~1.4–2.1 $M_\odot$ compressed into radii of ~10–13 km—provide the most extreme observable test of the Aether framework. Their central densities ($\sim 10^{17}$–10^{18} kg/m^3) exceed normal nuclear density by factors of 2–8, approaching the compression regime of the strong-force bridges described in Section 4.8. Three classes of neutron star observations have direct, quantitative connections to the theory's predictions.

3.1 Pulsar Glitches and Quantized Vortex Dynamics

Pulsars (rotating neutron stars) emit beams of radiation at precisely measured frequencies. Occasionally, a pulsar's rotation frequency increases abruptly by a fractional amount $\Delta\Omega/\Omega \sim 10^{-9}$–$10^{-6}$—a "glitch"—then relaxes back over timescales of hours to months. The Vela pulsar glitches approximately every three years; the 2016 event had a rise time under 12.6 seconds [25, 26].

The standard explanation, confirmed by detailed microphysical modeling [27], involves quantized vortex lines in a superfluid neutron component:

1. As the crust slows (electromagnetic braking), the interior superfluid retains its angular momentum through pinned quantized vortex lines.
2. Differential rotation builds stress between the pinned vortex array and the decelerating crust.
3. When the stress exceeds a critical threshold, vortex lines unpin in an avalanche, transferring angular momentum to the crust—a sudden spin-up.
4. The system relaxes as vortex-mediated mutual friction recouples the components.

In the Aether framework, this is not merely an analogy—it is a macroscopic manifestation of the same vortex physics that underlies the theory. The neutron star interior is a region where the Aether is compressed to $\sim 10^6$ times its ambient density, and the superfluid behavior attributed to neutron

Cooper pairs in the standard picture is, in this framework, the superfluid behavior of the compressed Aether itself. The quantized circulation $\kappa = n\,h/m_A$ that stabilizes electron vortices at ambient density operates at neutron star densities with the same topological protection—the vortex core size ξ shrinks (since $c_{\text{local}} = \sqrt{K_A/\rho}$ decreases with increasing ρ), but the quantization condition remains exact.

The glitch phenomenology provides specific observational constraints:

- **Mutual friction coefficients.** The glitch rise time (< 12.6 s for Vela) and the multi-timescale relaxation constrain the mutual friction parameter $\mathcal{B}$, which in the Aether model depends on the interaction between quantized vortex cores and the surrounding medium at extreme density.
- **Vortex pinning energies.** The glitch amplitude and recurrence interval constrain the pinning energy per vortex, which in the Aether framework is the energy required to detach a quantized circulation line from the local density structure of the crustal lattice.
- **Superfluid fraction.** Bayesian analysis of the 2016 Vela glitch indicates that approximately 70% of the interior neutrons participate in the superfluid component [27]. In Aether terms, this constrains the fraction of the compressed medium that maintains quantum coherence and quantized circulation at these extreme densities.

3.2 The Equation of State at Extreme Density

The mass–radius relationship of neutron stars is the most direct observational probe of the equation of state (EOS) of matter at supranuclear density. The NICER X-ray telescope has measured the radii of several pulsars by tracking X-ray hotspots on their rotating surfaces, constraining the radius of a $1.4\,M_\odot$ neutron star to approximately 12.3 km [28, 29].

The Aether framework predicts a specific EOS—the logarithmic equation of state:

$$P = K_A \ln\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{10}$$

derived from the requirement of constant bulk modulus K_A. At neutron star densities ($\rho \sim 10^{17}$–10^{18} kg/m^3), the compression ratio relative to ambient Aether is $\rho/\rho_A \sim 10^6$–10^7, approaching but not reaching the bridge compression ratio $X \approx 7 \times 10^7$. The logarithmic EOS predicts:

$$P_{\text{NS}} \approx K_A \ln\left(\frac{10^{17}}{10^{11}}\right) \approx 14\,K_A \tag{11}$$

With $K_A \sim 10^{28}$–10^{29} J/m^3, the longitudinal contribution alone gives $P_{\text{long}} \sim 10^{29}$–$10^{30}$ J/m^3—below the observed neutron star core pressures of $\sim 10^{34}$–10^{35} Pa by ~5 orders of magnitude. As discussed below, the spin stiffness $K_{\text{spin}} \propto \rho$ contributes an additional term that dominates at nuclear density, raising the effective pressure to the correct range.

The logarithmic EOS is qualitatively different from the polytropic and piecewise models typically used in nuclear astrophysics. Its constant bulk modulus produces a pressure that increases less steeply than a stiff polytrope ($P \propto \rho^{\gamma}$ with $\gamma > 2$) at moderate density, but never softens at high density either—the logarithm continues to rise without limit, though slowly. Whether this specific functional form is consistent with the observed mass–radius constraints from NICER and the tidal deformability $\tilde{\Lambda}$ measured from the binary neutron star merger GW170817 [30] is a quantitative test that could confirm or falsify the predicted EOS.

An important caveat: the logarithmic EOS with constant K_A has been confirmed experimentally only at compression ratios up to $n^2 \approx 6$ (via the Fresnel drag coefficient in diamond). Neutron star interiors probe compression ratios of 10^6–10^7, where the EOS has not been independently verified. Furthermore, the bridge analysis (Paper 5) identifies a second elastic modulus: the spin stiffness $K_{\text{spin}} \propto \rho$ governs transverse modes and grows with density, while the bulk modulus K_A (constant) governs longitudinal modes. At high compression X, the effective stiffness $K_{\text{eff}} \approx K_A + \alpha\, K_{\text{spin}} = K_A(1 + \alpha X)$ is dominated by the spin term, giving an effective speed of sound $c_{\text{eff}} \approx \sqrt{\alpha}\, c$ at nuclear density (where α is the spin–compression coupling strength). For $\alpha \approx 0.25$, $c_{\text{eff}} \approx 0.5\, c$—consistent with neutron star constraints. The tidal deformability $\tilde{\Lambda}$ from GW170817 therefore tests the *combined* bulk and spin stiffness, not K_A alone.

3.3 Deconfinement and Quark Cores

The central density of the most massive neutron stars ($\gtrsim 2\, M_\odot$) may exceed the QCD deconfinement threshold of ~5–10 times nuclear density. If so, the innermost core would contain deconfined quark matter—a quark–gluon plasma (QGP). Current multi-messenger observations (gravitational waves from mergers, NICER mass–radius constraints, and nuclear theory benchmarks) are actively probing whether such phase transitions occur inside neutron stars [31].

The Aether framework makes a specific prediction for what deconfinement means physically: the binding bridges (compressed Aether strands anchored by quark vortex endpoints) dissolve when thermal energy exceeds the elastic energy per bridge cross-section. In a neutron star core, this occurs when the temperature and density reach the deconfinement regime described in Section 1.

The bridge clarification developed in Section 4.8—that bridges are elastic compressions of the same superfluid phase, stabilized by quark vortex endpoints—leads to a distinctive prediction. When deconfinement occurs, the elastic compression energy stored in the bridges ($\varepsilon \approx K_A\, X$ per unit volume of bridge material) is released as the compressed Aether relaxes to local equilibrium. This energy release should produce an observable signature: a softening of the effective EOS at the deconfinement transition, potentially detectable as a kink in the mass–radius curve or a characteristic frequency in the gravitational wave spectrum of a neutron star merger.

The theory also predicts that the quark–hadron transition is *not* a sharp first-order phase transition (as some QCD models suggest) but a gradual crossover. In the Aether picture, deconfinement is the thermal disruption of individual bridge endpoints—a process that occurs progressively as temperature and density increase. Different bridges dissolve at different local conditions, producing a smooth transition rather than a discontinuous jump. Recent Bayesian analyses of neutron star data favor smooth crossover models over sharp phase transitions [31], consistent with this prediction.

4. Summary of Theoretical Properties

4.1 Properties of Black Holes

- The event horizon is a sonic horizon: the Aether inflow velocity $v_{\text{in}}(r) = \sqrt{2GM/r}$ reaches c at the Schwarzschild radius $r_s = 2GM/c^2$, preventing outward wave propagation. This is physically identical to a sonic horizon in fluid dynamics.

- Hawking radiation arises from asymmetric transient binding near the horizon: the restoring pressure pulse from pair annihilation is partially swept inward by the near-c inflow, producing a net outward energy flux. The temperature depends on the inflow velocity gradient at the horizon: $|dv_{\text{in}}/dr| = c^3/(4GM)$, giving $T \propto 1/M$.

- The Hawking effect and the Unruh effect are the same phenomenon in the Aether model: asymmetric transient binding caused by velocity gradients in the medium, whether from gravitational inflow or from acceleration.

- The photon sphere at $r = 3GM/c^2$ arises from the acoustic metric of the inflow bending reaching the strength required for circular photon orbits. The black hole shadow observed by the Event Horizon Telescope is the photon sphere's projection.

- Frame dragging (Lense–Thirring effect) is the co-rotation of Aether driven by the aggregate electron vortex circulation of a rotating mass. A rotating black hole creates an ergosphere where the dragged Aether flow exceeds what any stationary object can resist.

- The no-hair theorem follows from the sonic horizon: only mass (inflow profile), angular momentum (rotational flow), and charge (coherent axial-oscillation wave field) produce long-range Aether patterns that survive outside the horizon. All other information is swept inward.

- The singularity may be avoided if K_A increases at extreme compression, limiting the maximum achievable density to a finite value (potentially at or near the Planck density).

4.2 Properties of Neutron Stars

- Neutron star interiors are regions where Aether is compressed to $\sim 10^6$–10^7 times ambient density, approaching the bridge compression ratio $X \approx 7 \times 10^7$.
- Pulsar glitches are macroscopic manifestations of quantized vortex dynamics: the same circulation quantization $\kappa = nh/m_A$ that stabilizes electrons at ambient density operates in the extreme-density interior, with vortex unpinning avalanches producing sudden spin-ups.
- The logarithmic equation of state $P = K_A \ln(\rho/\rho_0) + P_0$, combined with the spin stiffness $K_{\text{spin}} \propto \rho$ at nuclear density, produces neutron star core pressures consistent with observations ($\sim 10^{34}$–10^{35} Pa) and makes specific, falsifiable predictions for the mass–radius relationship and tidal deformability.
- At sufficiently high core density and temperature, the Aether bridges confining quarks dissolve (deconfinement), producing a quark core. The theory predicts this transition is a gradual crossover (progressive bridge disruption) rather than a sharp first-order phase transition.
- The released elastic compression energy from bridge dissolution should produce a softening of the effective EOS at the deconfinement transition, potentially detectable in gravitational wave spectra from neutron star mergers.

4.3 Properties of Dark Energy and Dark Matter

- Dark energy is the thermal pressure of hot Aether in cosmic voids, pushing against the cold Aether regions near matter. Driven by accumulated radiated energy from stars with no mass formation to absorb it. Produces real expansion and real Doppler redshift, naturally explaining supernova time dilation.
- Dark matter effects arise from the two-fluid first-sound correction: the normal fraction increases the first-sound speed $c_1 > c$, and the outward gradient $dc_1^2/dr > 0$ provides additional centripetal acceleration through the acoustic metric—no additional mass required.
- The Aether thermal pressure model predicts that dark energy evolves over cosmic time—consistent with DESI observations of a time-varying equation of state at 2.8–4.2σ significance.
- Photons traversing voids encounter modified Aether properties (higher temperature, larger normal fraction), potentially amplifying the integrated Sachs–Wolfe effect beyond ΛCDM predictions.
- Void density profiles are predicted to be approximately universal when scaled by void radius, due to the invariance of K_A.

5. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **Accelerating expansion.** The observed acceleration of cosmic expansion is consistent with this theory's prediction of a positive feedback loop: matter radiates energy into voids, heating the Aether, increasing thermal pressure, and driving faster expansion.

2. **Supernova time dilation.** Distant supernova light curves are stretched proportional to their redshift. This is naturally produced by the real expansion mechanism of this theory (thermal pressure driving genuine recession), matching the predictions of standard expansion-based cosmology.

3. **Failed dark matter particle searches.** Extensive experimental programs (LUX [7], XENON [8], PandaX [9], and others) have failed to detect dark matter particles despite decades of effort. This is consistent with this theory's prediction that dark matter is not a particle but a medium property—the gravitating kinetic energy of the Vinen-regulated fountain counterflow in the two-fluid Aether.

4. **Black hole shadow observations.** The Event Horizon Telescope imaged the shadow of M87* [5] and Sagittarius A* [6], confirming the existence and size of the photon sphere to approximately 10% precision. Since the Aether model reproduces the weak-field bending angle exactly, the strong-field photon sphere and shadow size are expected to be consistent with these observations, subject to quantitative verification through full ray-tracing calculations.

5. **Acoustic Hawking radiation [3].** A Bose–Einstein condensate experiment created a sonic horizon in a flowing superfluid and observed thermal phonon emission at the predicted Hawking temperature. In the Aether model, this is not an analogy but the same physical mechanism: thermal radiation from asymmetric fluctuations at a sonic horizon in a flowing medium.

6. **Analog Hawking radiation in a Bose–Einstein condensate.** Steinhauer [21] created a sonic black hole in an ultracold rubidium condensate by engineering a flow exceeding the local speed of sound, and observed spontaneous emission of entangled phonon pairs at the acoustic horizon—the analog of Hawking radiation. In the Aether model, black holes are sonic horizons where Aether inflow velocity reaches c. The observation that a superfluid sonic horizon spontaneously produces quantum radiation confirms the mechanism by which this theory predicts Hawking emission.

7. **Rotating black hole geometry from a giant quantum vortex.** Svancara et al. [22] stabilized a giant quantum vortex in superfluid ^{4}He carrying thousands of circulation quanta and showed

that surface wave dynamics reproduce the curved spacetime geometry of a rotating (Kerr) black hole, including ergosphere signatures and ringdown physics. In the Aether model, gravity *is* Aether flow, and a black hole *is* a sonic horizon. The fact that a large vortex in a quantum superfluid spontaneously generates the geometry of a rotating black hole is a direct prediction of this framework.

8. **DESI evidence for evolving dark energy.** The Dark Energy Spectroscopic Instrument [14], using baryon acoustic oscillation measurements from over 14 million extragalactic objects, finds 2.8–4.2σ evidence that the dark energy equation of state varies with redshift— inconsistent with a cosmological constant and consistent with the Aether model's prediction that dark energy (thermal void pressure) evolves over cosmic time.

9. **Anomalous ISW signal from supervoids.** Galaxy surveys find that supervoids produce integrated Sachs–Wolfe temperature imprints approximately 5.2 ± 1.6 times stronger than ΛCDM predicts [20]. The Aether model's prediction of modified photon propagation through high-temperature void regions (larger normal fraction) could account for this amplification.

10. **Universal void density profiles.** When scaled by the void effective radius, observed void density profiles are approximately universal—independent of void size and cosmic epoch—consistent with a medium whose bulk modulus K_A is invariant everywhere.

11. **Pulsar glitches as quantized vortex avalanches.** The Vela pulsar's 2016 glitch [25, 26] exhibited a rise time under 12.6 seconds and multi-timescale relaxation, both explained by the sudden unpinning of quantized vortex lines in a superfluid interior. In the Aether model, these are quantized circulation lines ($\kappa = nh/m_A$) in extreme-density Aether—the same vortex physics that produces the Pauli exclusion principle and electron stability, observed at macroscopic scales.

12. **NICER mass–radius constraints.** X-ray observations of millisecond pulsars [28, 29] constrain neutron star radii to $\sim$12.3 km for $1.4\,M_\odot$ stars, directly testing the equation of state at supranuclear densities. The logarithmic EOS predicted by the Aether framework ($P = K_A \ln(\rho/\rho_0) + P_0$) combined with the spin stiffness $K_{\text{spin}} \propto \rho$, produces neutron star core pressures consistent with observations and can be quantitatively compared to these constraints.

13. **Binary neutron star merger GW170817.** The tidal deformability measured from this merger [30] constrains the stiffness of the EOS at 2–6 times nuclear density. The Aether prediction of constant K_A (logarithmic EOS) makes a specific, falsifiable prediction for the tidal deformability parameter $\tilde{\Lambda}$.

14. **Galaxy rotation curves.** Galaxies exhibit flat rotation curves requiring $\rho_{\text{DM}} \propto 1/r^2$ halos with cored central profiles. The two-fluid first-sound correction (Section 1.3) naturally produces

isothermal $1/r^2$ profiles with cores, matching observations without invoking a dark matter particle.

15. **The Hubble tension.** Early-universe (CMB) and late-universe (distance ladder) measurements of H_0 disagree at ~5σ. The Aether framework resolves this through environment-dependent thermal pressure: voids, which dominate the local distance ladder path, have higher Aether temperature and therefore faster local expansion than the cosmic mean sampled by the CMB.

6. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Void temperature and expansion rate.** The theory predicts a direct relationship between void temperature (or thermal energy content) and expansion rate. Measurements correlating the thermal properties of cosmic voids with local expansion rates could test whether thermal pressure is the driver of dark energy.

2. **Two-fluid first-sound dark matter: open calculations.** The first-sound correction mechanism (Section 1.3, Eq. 2) has the correct sign and produces naturally cored profiles, but its magnitude depends on the Aether's excitation spectrum. The critical open calculations are: (a) determine the Aether's excitation spectrum beyond the phonon regime—does it have roton-like intermediate-wavelength modes that break the temperature-independence of the phonon gas correction? For a pure phonon gas, the first-sound correction is $c^2/3$ (constant, producing no spatial gradient); non-phonon excitations are needed to create the temperature-dependent variation required for dark matter ($\Delta c_1^2/c^2 \sim 10^{-6}$ for galaxies; the ideal BEC gives ~10^{-4}, ~$50\times$ too large); (b) derive the three-dimensional temperature profile $T(r,z)$ from first principles (stellar heating vs. binding cooling vs. conduction); (c) determine whether the resulting $c_1(r)$ profile produces flat rotation curves; (d) calculate the CMB power spectrum from the cosmological superfluid phase transition; (e) verify Bullet Cluster consistency.

3. **Quantitative Aether-enhanced ISW effect.** The Aether framework predicts that photons traversing cosmic voids encounter modified medium properties (higher temperature, larger normal fraction). Can a quantitative calculation of the resulting temperature imprint reproduce the factor-of-five amplification of the ISW signal observed in supervoids [20]? This requires modeling the void temperature and density profiles within the three-property framework.

4. **Dark energy equation of state from the two-fluid model.** The normal-component bulk viscosity $\zeta_n \sim 3\times10^7$ Pa s (Eq. 1) and the cosmic star formation rate history together determine the heating–dilution balance that governs the dark energy equation of state $w(z)$. Can the

predicted w_0–w_a match the DESI observation of a phantom-to-quintessence crossing at $z \approx 0.5$? This is a specific numerical calculation using $\dot{\varepsilon}_{\rm heat} = L_*(z) + 9\,\zeta_n\,H^2(z)$ and $\dot{\varepsilon}_{\rm dilute} = 3H\,P_{\rm th}$.

5. **Cosmic coincidence from shared temperature structure.** Both dark matter (the first-sound correction $\Delta c_1^2/c^2$) and dark energy (the thermal pressure $P_{\rm th}$) arise from the same temperature field $T(\mathbf{x})$. Does the observed ratio $\Omega_{\rm DM}/\Omega_\Lambda \approx 0.4$ follow naturally from the temperature profile shape, without fine-tuning? If so, this would resolve the cosmic coincidence problem.

6. **Environment-dependent expansion and dark matter correlation.** The two-fluid framework predicts that the local expansion rate and the local dark matter fraction are correlated (both increase with the normal fraction, which increases with temperature). Upcoming DESI and Euclid void catalogs could test whether galaxies in hotter environments (near large voids) show both larger "dark matter" halos and faster local expansion.

7. **Bulk viscosity ζ_n and observational constraints.** The predicted $\zeta_n \sim 10^6$–3×10^7 Pa s gives a kinematic bulk viscosity $\nu_\zeta = \zeta_n/\rho_n \sim 10^{-4}$–$10^{-3}$ m^2/s (tiny, because $\rho_n \sim 10^{10}$ kg/m^3). The viscous damping scale is therefore submicroscopic—the normal fraction dissipates energy from cosmic expansion without damping astrophysical density perturbations. In superfluids, ζ arises from three-phonon relaxation and diverges near T_c; can ζ_n be independently constrained from the Aether's excitation spectrum or from its effect on second-sound attenuation?

8. **Void temperature ratio and the Hubble tension.** The quantitative analysis of Section 1.7 shows that matching the observed Hubble tension requires a void thermal pressure excess of $\eta \approx 0.22$ (Eq. 6), corresponding to a void-to-wall temperature ratio $T_{\rm void}/T_{\rm wall} \approx 2.5$ (Eq. 7). Can this ratio be derived from a thermodynamic model of void heating and wall cooling over cosmological time? Specifically: (a) does the integrated radiative energy deposited into voids, combined with energy absorbed by mass formation in walls, produce the required factor of $\sim$2.5? (b) Does the predicted H_0 anisotropy pattern match the dipole and quadrupole structure seen in recent anisotropy analyses of the distance ladder? (c) Can the void abundance predicted by the thermal pressure mechanism be tested against upcoming DESI and Euclid void catalogs?

9. **Two-fluid Aether: critical temperature and superfluid fraction.** The two-fluid dark matter mechanism requires a finite T_c and a continuously varying superfluid fraction. For $m_A \approx 500$–850 MeV/c^2 (Paper 7), the ideal BEC critical temperature is $T_c \sim 10^{10}$–10^{11} K, placing the cosmological phase transition well before nucleosynthesis. Key questions: (a) What is the superfluid fraction in cosmic voids today? (b) Does the anomalous specific heat near T_c (the lambda anomaly) thermodynamically stabilize the halo boundary, creating sharper edges than NFW profiles? (c) Is T_c high enough that the entire present-day universe remains below it ($T_{\rm void} < T_c$), ensuring the condensate exists everywhere and the universe is transparent?

10. **Photon dispersion from the normal fraction in voids.** At macroscopic wavelengths, photons propagate at the two-fluid first-sound speed c_1 (via spin-orbit coupling), which is achromatic.

However, at wavelengths approaching the spin healing length ξ_s, the spin-wave dispersion relation acquires corrections. Because the normal fraction is larger in voids, the spin-wave–normal-fluid coupling (suppressed by $P_{\text{th}}/K_A \sim 10^{-38}$) should produce a residual *differential* dispersion: photons traversing supervoids should exhibit slightly more dispersion than photons traveling through matter-rich regions (smaller normal fraction). The Fermi LAT constrains Lorentz-violating dispersion to energy scales above $7.6\,E_{\text{Planck}}$ [13], but a differential void-versus-filament comparison could reveal or constrain the void superfluid fraction.

11. **Matter formation at the cosmological phase transition.** During the Aether's superfluid phase transition ($T_c \sim 10^9$ K, $t \sim 1$ s), the Kibble–Zurek mechanism [23, 24] predicts copious creation of quantized vortices (= particles) at the transition front. The resulting matter concentrations produce gravitational wells that seed structure formation *before* any galaxies exist, potentially resolving the early-universe dark matter requirement. Does the spatial spectrum of Kibble–Zurek defects from the Aether transition match the CMB power spectrum?

12. **Second sound in the cosmic Aether.** The two-fluid model predicts a second propagation mode: temperature waves (second sound) at speed $c_2 \ll c$. Sudden heating events (supernovae, galaxy mergers, quasar ignition) would launch second-sound pulses through the Aether. These temperature waves would appear as propagating oscillations in the local dark energy density. Could second sound contribute to the baryon acoustic oscillation signal or other large-scale features? The speed c_2 is determined by the superfluid fraction and the Aether's thermodynamic properties—both calculable in principle.

13. **Critical velocity and quantum turbulence near compact objects.** The gravitational inflow $v_{\text{in}} = \sqrt{2GM/r}$ exceeds the Landau critical velocity $v_c = \Delta/p_{\text{roton}}$ at some radius near compact objects, nucleating quantized vortices and driving the Aether into a quantum turbulent state. For neutron stars, the resulting vortex dynamics may connect to the observed pulsar glitch mechanism (vortex unpinning events). For black holes, the region between v_c and the sonic horizon is a quantum turbulent shell whose properties affect the Hawking temperature and gravitational wave ringdown signatures.

7. Proposed Experiments

The following experiments target specific predictions relevant to this paper:

1. **Neutron star gravitational redshift as a strong-field test.** General relativity predicts the surface gravitational redshift of a neutron star as $z = (1 - 2GM/Rc^2)^{-1/2} - 1$. The Aether model's logarithmic equation of state produces a nonlinear density profile in the strong-field regime ($\delta\rho/\rho_0 \gtrsim 0.1$) that may yield a different redshift for the same mass and radius.

NICER pulse-profile modeling now provides simultaneous M and R measurements for several millisecond pulsars; combining these with spectroscopic redshift measurements from X-ray absorption lines (e.g., 1E 1207.4−5209, EXO 0748−676) would allow a direct comparison between the GR and Aether predictions. A statistically significant discrepancy at the ~ 5–10% level would distinguish the two frameworks in the strong-field regime.

2. **CMB frame and Aether rest frame coincidence.** Special relativity asserts that no preferred rest frame exists; the CMB dipole (our motion at ~ 370 km/s relative to the cosmic microwave background) is merely a Doppler effect with no deeper significance. The Aether model predicts that the medium *does* have a rest frame. If the Aether rest frame coincides with the CMB rest frame (natural if CMB photons are Aether excitations), then all precision tests of Lorentz invariance should show residual anisotropies aligned with the CMB dipole direction. If they do *not* coincide, a "dark flow"—a systematic velocity of the Aether relative to the CMB—would produce unexplained large-scale anisotropies in galaxy surveys. Current large-scale structure surveys (DESI, Euclid) and next-generation CMB experiments (CMB-S4) have the sensitivity to detect or rule out bulk flows at the ~ 100 km/s level, constraining or confirming the Aether frame hypothesis.

For all proposed experiments, see Paper 1: General Overview [40].

8. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [40].*

8.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

Symbol	Name	Constrained range	Primary constraint
ρ_A	Ambient density	10^{11}–10^{12} kg/m^3	Electron vortex energy
K_A	Bulk modulus	10^{28}–10^{29} J/m^3	$K_A = \rho_A\, c^2$
m_A	Aether quantum mass	500–850 MeV/c^2	Flux tube width / glueball mass
ξ	Healing length	0.17–0.28 fm	Bridge diameter (lattice QCD)

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \tag{12}$$

$$\xi = \frac{\hbar}{m_A\, c\, \sqrt{2}} \tag{13}$$

leaving two free Aether parameters (e.g., ρ_A and m_A) from which all others follow. (A third parameter, ϵ, enters through gravity; see below.)

8.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section 4):

$$P = K_A \ln\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{14}$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so c does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. This equation of state guarantees that K_A is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6}\,c$ to $0.9994\,c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to ~6:1.

8.3 Cosmology

Dark energy. Thermal pressure of hot Aether in cosmic voids (density essentially uniform, $\delta\rho_A/\rho_A \sim 10^{-4}$; temperature is the cosmologically relevant variable):

$$P_{\text{void}} = K_A \ln\left(\frac{\rho_{\text{void}}}{\rho_0}\right) + P_{\text{th}}(T_{\text{void}}) \tag{15}$$

where $P_{\text{th}}(T)$ is the thermal pressure from the Aether's excitation spectrum ($P_{\text{th}} \propto T$ for an ideal gas, $P_{\text{th}} = \frac{1}{3}a_R T^4$ for a radiation-dominated spectrum; the actual form is determined by the same excitation spectrum that sets the dark matter magnitude and the bulk viscosity). The theory predicts dark energy evolves over cosmic time (consistent with DESI observations at 2.8–4.2σ).

Dark matter. The Landau two-fluid first-sound correction: the normal fraction increases c_1 above c, and the outward gradient $dc_1^2/dr > 0$ provides additional centripetal acceleration through the acoustic metric (no additional mass required). Rotation curves and gravitational lensing are automatically consistent (same acoustic metric). The Milky Way halo requires $\Delta c_1^2/c^2 \approx 6 \times 10^{-6}$; the magnitude depends on the Aether's excitation spectrum. Naturally produces cored profiles.

Vacuum catastrophe resolution. The QFT vacuum energy density $\sim 10^{113}$ J/m^3 is reinterpreted as $K_A \sim 10^{28}$ J/m^3 (the actual ground-state energy density of the Aether), resolving the 10^{85}-fold discrepancy between the predicted vacuum energy and the Aether ground-state energy density (compared to the 10^{120}-fold discrepancy with the observed cosmological constant).

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